

Multiple Choice (10 marks)

1. Statement 1| ImageNet has images of various resolutions. Statement 2| Caltech-101 has more images than ImageNet.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 3 & 2 & -9 \\ -6 & -4 & 18 \\ 12 & 8 & -36 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
4. After applying a regularization penalty in linear regression, you find that some of the coefficients of w are zeroed out. Which of the following penalties might have been used?
 - (a) L 0 norm
 - (b) L 1 norm
 - (c) L 2 norm
 - (d) either (a) or (b)
5. Averaging the output of multiple decision trees helps ___.
 - (a) Increase bias
 - (b) Decrease bias
 - (c) Increase variance
 - (d) Decrease variance

True / False (10 marks)

Answer True or False

6. True or False: Residual connections are commonly utilized in both ResNets and Transformers to facilitate training of deep networks.
7. True or False: The joint probability $P(A, B, C)$ can be expressed as $P(A, B | C) * P(C)$.
8. True or False: Underfitting occurs when a model fails to capture the underlying patterns in the training data and cannot generalize to new data.
9. True or False: The method used to learn weights, such as matrix inversion or gradient descent, primarily affects the trade-off between underfitting and overfitting.
10. True or False: Clustering techniques are effective for detecting fraudulent credit card transactions by grouping similar behaviors.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the validity of the statements regarding entropy in joint distributions and the effects of moralization on directed graphs. Analyze their implications in machine learning contexts.
12. Discuss the assumptions of the Naive Bayes classifier, specifically addressing the implications of attribute independence given the class value. Why is this assumption critical?

End of Paper